

Continuity equation

$$\frac{V_i}{c_{i,j}^2 \Delta t} (p_i^{n+1} - p_i^n) = \sum_{j=1}^k G_{i,j}^{n+1} - G_{ext,i}^{n+1}$$

Flow equation

LINEARIZED EQUATION

$$\begin{aligned} \Delta P_j^{(k+1),n+1} - \left(2R_{f_j} \left| G_j^{(k),n+1} \right| + R_{I_j} \right) G_j^{(k+1),n+1} \\ = - R_{f_j} \left| G_j^{(k),n+1} \right| G_j^{(k),n+1} - R_{I_j} G_j^n \end{aligned}$$

Coupled
fluid-dynamic problem